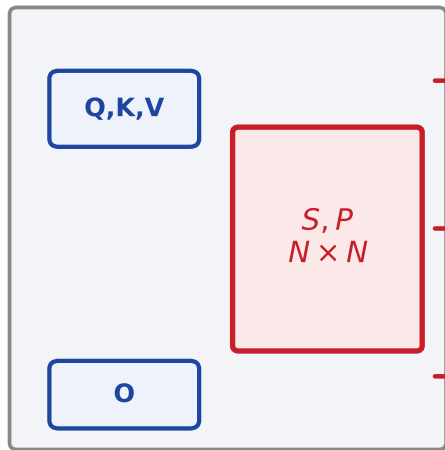


GPU HBM (slow, large)



several full $N \times N$ round-trips to slow HBM

1. $S = QK^T$

write $N \times N$

2. $P = \text{softmax}(S)$

read + write $N \times N$

3. $O = PV$

read $N \times N$, write O

softmax / mask / dropout are cheap in FLOPs but memory-bound

schematic